Matched Specialization, Different Adaptation: Second-Moment Structure in Shared Representations

Toru Takahashi
Human Informatics and Systems Lab, Doshisha University
Linked Open Data Initiative, NPO
Keio Research Institute at SFC
Stroly Inc.

Abstract

Can two learners with similar measured specialization adapt differently to the same new learning condition? We study this question in small shared-representation networks whose formation histories and optimizer states can be controlled. Different observation and loss-weighting histories produce equivalent mean reconstruction measurements. Under a common intervention favoring the other target, the more precisely readable target switches within ten updates in 67.5% of acquisition-formed checkpoints and 0% of update-formed checkpoints. Resetting Adam reduces the between-route difference in reallocation to within a pre-specified equivalence margin. In an independent component-exchange experiment, swapping the second moment reverses which history is more susceptible. Further interventions separate its mean adaptive scaling from its coordinate-wise pattern: equalizing the mean scale reduces the difference, while exchanging the pattern at that common scale changes its sign. Fresh Adam and calibrated SGD reveal smaller delayed differences, while sustained common learning brings allocation endpoints close by 500 updates. Thus measured specialization can conceal a reversible difference in short-horizon adaptation, with second-moment structure acting as a causal modifier. The matched comparison extends to three Gaussian targets at two bottleneck capacities. A three-room sensor-data extension retains component effects but fails endpoint matching, delimiting the evidence for concealed susceptibility.

1 Introduction

A shared representation is an internal code used by several prediction tasks. With limited capacity, information useful for one task may remain easier to recover than information useful for another. We call this inequality in readout precision *representational specialization*. Such biases are well established: feature complexity, learning order, and competition among information sources can change what a network represents (Lampinen et al., 2024; Pezeshki et al., 2021; Huang et al., 2022). We ask what a measured specialization tells us about the learner's response when learning conditions change.

This question matters whenever a model continues learning after its performance has been evaluated. Two models may currently support similar predictions but require different amounts of learning to reallocate their shared features. Training history can affect later plasticity, the ability to learn from new information, through both model parameters and optimizer state (Achille et al., 2019; Lyle et al., 2023; Dohare et al., 2024). The experimental challenge is to distinguish a difference in current measured ability from a difference in subsequent response, and then identify a persistent component that changes that response.

We construct this comparison using two routes to specialization. One route observes a target more precisely while weighting reconstruction errors equally; the other balances precise observations while placing more weight on that target's error. Pilot calibration makes selected mean endpoint measurements equivalent

[*] Generative AI assisted experiment design, implementation, analysis, literature work, and writing; see the AI assistance statement.

within fixed margins. We then apply the same intervention favoring the other target. If those measurements adequately characterize adaptation under this intervention, the routes should respond similarly. If inherited optimizer state contributes to an otherwise hidden response difference, resetting or exchanging that state should change the contrast while leaving each recipient's model weights fixed at intervention onset.

The results support the second explanation for a large early response difference. Adam's second moment, its running estimate of squared gradients, rescales updates separately for each parameter. Exchanging it reverses which formation route changes more. Equalizing its mean inverse scaling leaves a response difference, and exchanging its coordinate-wise pattern at a common mean scale changes the sign. These interventions identify how the organization of adaptive scaling modifies reallocation in the recipient network. Longer trajectories then show why the observation horizon matters: smaller route differences emerge even after the inherited Adam state is removed, and prolonged common learning brings the measured allocation endpoints close.

Our empirical contribution has three connected parts: a comparison of adaptation after matching selected reconstruction measurements; component and scale/pattern interventions that change the short response difference; and a temporal comparison that separates this large inherited-state transient from smaller delayed differences. We use *latent susceptibility* to name the response difference concealed by those endpoint measurements. Small artificial learners allow the paired histories, saved model states, and optimizer components needed for these tests. The matched comparison covers two- and three-target Gaussian systems. A natural sensor-data extension tests which parts survive real cross-target and temporal structure; complementary formation and maintenance experiments locate these results within the wider process of specialization.

2 Relation to training history and optimizer memory

Similar measured performance can coexist with important differences between predictors, as studied under underspecification (D'Amour et al., 2022). Representation-similarity measures examine another aspect of their endpoints (Kornblith et al., 2019). Our comparison concerns the response to further training: selected reconstruction measurements are matched before a common intervention. This connects the study to plasticity research, where early experience and training dynamics change later learning (Achille et al., 2019; Jastrzebski et al., 2020; Dohare et al., 2024).

Optimizer-mediated history effects have direct precedents. Lyle et al. (2023) relates plasticity to loss geometry and shows how unequal adaptation rates of Adam's moment estimates can destabilize training after task changes. Asadi et al. (2023) evaluates optimizer resets after changing reinforcement-learning objectives. Ellis et al. (2024) analyzes update-size transients after gradient-scale changes and introduces relative Adam timesteps. These results make inherited adaptive scaling a concrete explanation to test here. Our question is whether average scaling suffices to account for a route difference in *representational reallocation*, or whether the coordinate-wise pattern also changes it.

Direct state transplantation has also been used to study hidden training effects. In a recent preprint, Xu (2026) identifies a first-moment carrier of subliminal transfer: an intervention invisible in immediate outputs produces differences during subsequent training. Sevetlidis & Pavlidis (2026) organizes historical carriers and proposes state-based causal measurements. The present contribution is a specific empirical decomposition of shared-representation adaptation. It combines matched reconstruction measurements, second-moment scale/pattern interventions, and a common continuation under inherited Adam, fresh Adam, and SGD. Table 1 summarizes the closest comparisons.

These studies establish why training state matters, while the measured outcomes differ. A change in specialization under a ten-update pulse is neither an optimization-performance improvement nor a measure of accumulated trait transfer. The interval for the change in response contrast under first-moment exchange lies within the pre-specified equivalence margin at this horizon. The longer trajectories test the duration of reallocation effects under a common opposing task. Transfer of learned attributes to a distinct downstream task remains an open comparison.

Table 1: Closest prior comparisons and the question tested here. The interventions and measured outcomes specify how the comparisons relate.

Prior work	Established comparison	Present comparison
Lyle et al. (2023)	Plasticity, loss geometry, and optimizer instability under changing tasks	Reallocation response after matching selected reconstruction measurements
Asadi et al. (2023); Ellis et al. (2024)	Resetting state or time; update-size effects of changed objectives or gradient scales	Component exchange and coordinate pattern at matched mean inverse scaling
Xu (2026)	First-moment transplantation and later transfer, including immediate-output invariance	Second-moment structure and the time course of specialization response

3 A controlled comparison of specialization and adaptation

3.1 Two formation routes through one shared bottleneck

The central learner reconstructs two three-dimensional signals, A and B, from noisy observations. We generate standard Gaussian A and independent Gaussian ϵ, and set $B = \rho A + \sqrt{1-\rho^2}\epsilon$ with $\rho = 0.35$. A seed-specific orthogonal transformation mixes the observations. A $6 \to 32 \to 2$ encoder, with tanh after both layers, maps them to a two-number code shared by two linear reconstruction heads. Training targets are the noise-free signals. This bottleneck makes the allocation of readable information observable.

We manipulate information quality and error weighting separately. *Acquisition precision* changes target-specific input noise before mixing; the high- and low-precision standard deviations are 0.05 and 1.20. Dividing each noisy channel by $\sqrt{1+\sigma^2}$ normalizes its variance. *Update allocation* changes the reconstruction-loss weights:

$$L_t = w_{A,t}\ell_{A,t} + w_{B,t}\ell_{B,t}, \qquad w_{A,t} + w_{B,t} = 1. \tag{1}$$

Equal weights give the errors equal explicit importance; their gradients can still differ. Observation quality therefore enters learning through a different intervention from changing loss weights. Related work studies active acquisition and task weighting (Shim et al., 2018; Chen et al., 2022; 2018; Yu et al., 2020); the present design crosses these controls in one shared representation.

After 100 balanced initialization updates, a nine-condition acquisition-by-weighting comparison runs for 500 updates to test which target each control favors. Balanced acquisition allocates exactly 250 high-precision updates to each target. For the matched comparison, the acquisition route observes the favored target precisely with equal loss weights; the update route balances acquisition and assigns the favored target weight 0.80. Route strengths are calibrated on pilot seeds and then fixed for confirmation. Full formation and redirection controls appear in Appendix B.

3.2 Common measurements and paired inference

To measure readable information, we freeze the encoder, fit fresh linear readouts on unused probe data, and evaluate on separate held-out data. Both targets receive common high-precision inputs. Let D_A, D_B be the variance-normalized mean squared errors. The allocation index

$$s = \frac{D_B - D_A}{D_A + D_B} \tag{2}$$

is positive when A is more precisely readable. We align its sign to the originally favored target. A decrease then measures reallocation toward the other target, and crossing zero changes which target is more readable. We also retain both errors, their sum, and allocation under a common low-precision probe. Thus relative advantage is evaluated alongside absolute reconstruction quality.

Each seed supplies a common ancestor and paired input/noise sequences. Conditions branch with the same initial model and, unless manipulated, optimizer state. Pilot and confirmatory seeds are disjoint;

Table 2: Route difference in specialization lost after ten opposing updates. Brackets give 95% seed-bootstrap intervals. All inherited-state intervals lie above +0.05, and all reset intervals lie inside ±0.02, meeting their respective pre-specified criteria.

Pulse	Adam inherited	Adam reset
Acquisition only	0.1668 [0.1601, 0.1736]	0.0011 [−0.0010, 0.0035]
Update only	0.1510 [0.1456, 0.1562]	0.0120 [0.0093, 0.0149]
Combined	0.1951 [0.1880, 0.2021]	0.0103 [0.0079, 0.0129]

each principal comparison uses 40 confirmatory seeds. Mirrored target labels are averaged within seed for inference. We report 95% seed-bootstrap intervals. Endpoint equivalence requires the interval for the mean paired difference to lie inside each pre-specified margin; it is not an assertion that every pair has identical measurements. All confirmatory seeds are retained. Model parameters and representation geometry may differ after formation.

Before confirmatory runs, each plan fixes the largest acceptable mean differences in allocation-index or normalized-error units. Appendix A lists these matching tolerances and response margins, together with the separate pilot stages.

The core evidence comprises the endpoint-matching experiment (IV-D), independent component exchanges and a within-route control (IV-E), and a prospective scale/pattern and time-course follow-up on 40 further seeds. Supporting experiments address formation (IV-B), redirection (IV-A/C), maintenance (II/III), and three-target formation (V-A). The three-target adaptation extension (V-B) and a natural sensor-data comparison (V-C) test the scope of the component effects with new seeds. Appendix A gives the exact interventions and full estimates, and Appendix I separates pre-specified tests, exploratory analyses, and failed candidates.

4 Matched specialization conceals a short-horizon response difference

Both controls can select specialization: under equal loss weights, precise acquisition of A or B produces mean allocation indices of 0.3385 and −0.3338 (Figure 1a), while sufficiently strong opposing update weights can overcome this acquisition advantage.

The two formation routes reach similar measured specializations. All five endpoint criteria pass: aligned s, target and non-target errors, total error, and low-precision aligned s. The mean aligned-s difference is −0.0064 ($[-0.0098, -0.0032]$), inside ±0.02. Mean integrated responses over 200 neutral continuation updates, with balanced observation and equal weights, are also equivalent with either inherited or fresh Adam. Matching therefore covers selected static measurements and one neutral-continuation summary, while leaving response to an opposing condition to be tested.

Each formed learner is copied into three ten-update opposing interventions. An acquisition-only pulse observes the previously less readable target precisely with equal loss weights. An update-only pulse balances acquisition and gives that target weight 0.80. A combined pulse observes it precisely and assigns it weight 0.85. Each pulse is tested with inherited or freshly initialized Adam, keeping the recipient model weights unchanged at the branch point. The response is the decrease in aligned s; the route contrast is acquisition-formed minus update-formed decrease.

With inherited Adam, the acquisition-formed route moves further under every pulse (Table 2). Under the combined pulse, 67.5% of its checkpoints cross zero within ten updates, compared with 0% of the update-formed checkpoints (Figure 1b). The continuous route difference is 0.1951. Resetting Adam reduces it to 0.0103, within the pre-specified ±0.02 margin. The other pulses show the same reduction. Thus inherited optimizer state strongly modifies a response difference that selected endpoint measurements and neutral continuation did not reveal.

This result applies to the two-target network and the ten-update pulses. A separate linear measure of representation similarity (CKA) averages 0.927, and the model parameters are not identical. Resetting Adam leaves those model differences intact. The small reset contrast therefore identifies a limited response

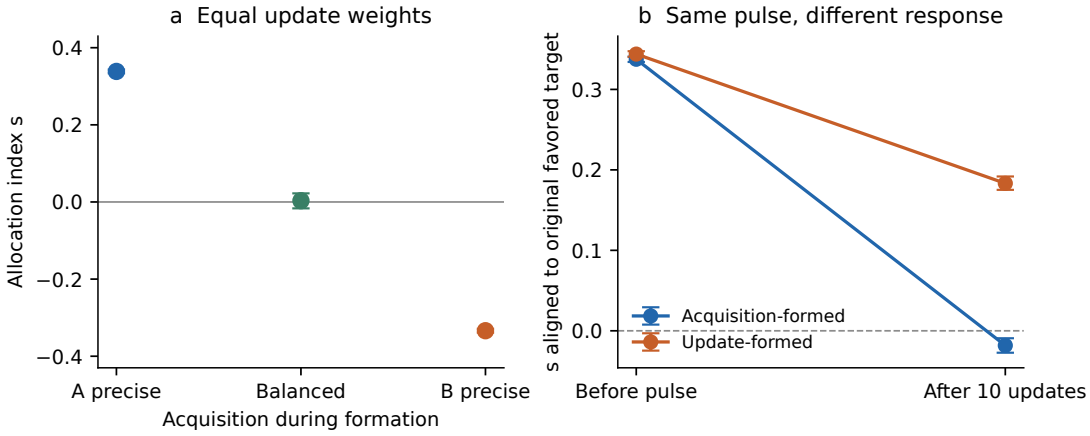


Figure 1: A common measured allocation can conceal different adaptation. (a) Acquisition selects specialization with equal loss weights. (b) The matched formation routes receive the same combined opposing pulse. Both move toward the other target, but the acquisition-formed route moves further. Points and intervals summarize 40 seeds; mirrored labels in (b) are averaged within seed. Lines connect endpoints, rather than showing the intervening trajectory.

equivalence at this horizon, rather than identical representations. The zero-crossing rates likewise describe the brief pulse; Section 6 examines longer common learning.

An exploratory check across the three existing 40-seed cohorts asks whether residual endpoint differences predict the response contrast within seeds. Linear models using target error, other-target error, and low-precision allocation provide little improvement over predicting the training-fold mean. This supports the limited predictive value of these measured endpoint features; Appendix J reports the held-out errors and conditional estimates.

5 Second-moment structure causally modifies reallocation

5.1 Component exchange reverses the route contrast

Adam retains a signed-gradient average m and a squared-gradient average v (Kingma & Ba, 2015). The latter sets a separate adaptive scale for each parameter. To test these components directly, we exchange m or v between route endpoints while holding each recipient's weights and optimizer step count fixed. Corresponding parameter coordinates are available because both routes descend from the same ancestor.

In an independent 40-seed comparison, exchanging m leaves the mean response contrast nearly unchanged, with the change inside the pre-specified margin. Exchanging v changes the route contrast from $+0.1968$ to -0.1265 (Figure 2a). Both estimates' intervals exclude zero in opposite directions (Appendix A). The recipient's second-moment state therefore changes which route is more susceptible. This intervention identifies a causal modifier acting together with the recipient weights and subsequent gradients; it measures the combined ten-update response.

5.2 Average scaling accounts for only part of the short response difference

The exchange raises a more specific question: does one route simply receive a larger average adaptive scale? In a prospective follow-up on 40 new seeds, the inherited-state contrast is 0.1907, reproducing the earlier 0.1951 and 0.1968 estimates. We define $\bar{a}_v$ as the mean of Adam's bias-corrected inverse-second-moment scaling terms across parameters. This quantity describes the scaling supplied by v; an actual update also uses the first moment and current gradient.

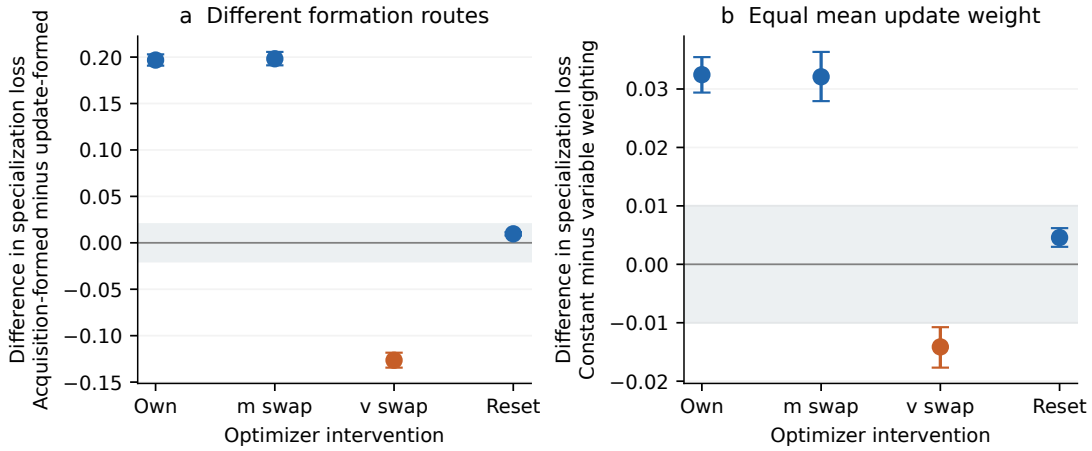


Figure 2: Second-moment exchange reverses two short-horizon response contrasts. (a) Acquisition-formed minus update-formed specialization loss. (b) Constant-minus-variable weight-history contrast at fixed acquisition and mean loss weight. Shaded bands show the respective pre-specified equivalence margins (± 0.02, ± 0.01). Intervals are paired seed-bootstrap intervals; each comparison uses 40 disjoint confirmatory seeds.

Equalizing $\bar{a}_v$ while preserving each route's coordinate pattern reduces the response difference from 0.1907 to 0.0875. Replacing the pattern by a scalar value gives 0.1251; combining scalarization with a common scale gives 0.0395 (Figure 3a). Exchanging the other route's pattern while holding $\bar{a}_v$ common changes the difference to -0.0110. Average scaling thus accounts for part of the response contrast, while coordinate pattern also changes the recipient's response at that common mean. This comparison equalizes a mean inverse-scaling quantity, not the actual update norms throughout the pulse. The post-hoc factorial recombination remains exploratory in Appendix A.

5.3 The effect also occurs within one formation-route category

A within-route control holds acquisition, mini-batches, update count, and mean target loss weight fixed. One schedule uses weight 0.80 throughout 500 updates; the other uses 1.00 and 0.60 for 250 updates each in randomized order. The endpoint criteria again pass. The constant schedule then moves further under the common pulse by 0.0324 ($[0.0294, 0.0354]$). Exchanging v reverses the contrast to -0.0141, while resetting Adam reduces it to 0.0046, inside ± 0.01 (Figure 2b). Different acquisition-versus-update labels are therefore unnecessary for this response pattern: equal average loss weights can conceal different optimizer states and later responses. Variability and realized sequence change jointly in this control.

5.4 Realized parameter movement links scaling to response

How does adaptive scaling relate to movement during the pulse? An exploratory analysis of the fixed follow-up trajectories measures cumulative path length by summing the distance moved by each update (its Euclidean norm). Over ten updates with inherited Adam, the acquisition-to-update route ratio is 0.987 ($[0.978, 0.996]$) for cumulative gradient norm and 1.872 ($[1.844, 1.902]$) for parameter-path length. Thus a gradient-norm ratio close to one accompanies substantially greater parameter movement in the acquisition-formed route.

Across the same 40 seeds, the between-route path-length difference correlates 0.806 with the reallocation-response difference. Holding out each seed in turn, single-variable linear models reduce mean squared error by 28% using the pre-intervention $\bar{a}_v$ ratio and 62% using the realized path-length difference, relative to predicting the mean response difference of the other 39 seeds. Realized movement thus provides a closer reconstruction of the response difference in this exploratory comparison.

A movement-controlled intervention on the 40 saved follow-up seed pairs tests whether unequal global path lengths are sufficient to explain the contrast. At every update, it rescales the four proposed vectors (two

routes, with own or exchanged v) to their common geometric-mean norm while preserving each direction. The route contrast is then 0.0303 with own v and 0.0072 with exchanged v; their paired difference is 0.0231. Thus v still alters reallocation under equal global movement. This controlled learning rule changes the trajectories and does not estimate a natural-Adam mediation fraction (Appendix J).

Layer-restricted exchanges locate a substantial part of this effect in the encoder. Starting from the original contrast 0.1907, exchanging only first-layer v leaves 0.0115, exchanging only second-layer v leaves 0.0611, and exchanging both heads' v leaves 0.1911. These interventions retain the recipient weights and all other optimizer components. Their non-additive effects localize effective interventions, rather than assigning independent shares of the full contrast.

5.5 Which parts of the comparison transfer across settings?

A fixed-protocol replication uses 32 new paired seeds at each of four settings: the baseline, hidden width 64, ReLU in place of tanh, and independent targets. All four satisfy the five endpoint tolerances. Each shows a positive inherited-state contrast, reversal under v exchange, and an exchange effect at common $\bar{a}_v$. Reset equivalence is more conditional: at width 64 the reset contrast is 0.0249, with its interval above 0.02. The replicated component effect therefore transfers more broadly than the claim that resetting Adam makes the ten-update responses equivalent. Appendix J gives every condition, including the failed reset criterion.

Varying β_2 and formation duration further bounds this transfer. Larger β_2 produces larger inherited-state contrasts in both tested duration groups, but v exchange does not reverse every contrast. Common-scale exchange effects remain positive throughout the six-setting grid. Thus changing second-moment dynamics alters the component effect, while reversal and reset equivalence depend on the training regime.

5.6 Three targets preserve the matched comparison; natural data delimit it

Does the component effect depend on having only one alternative target? We form acquisition- and update-driven histories for three symmetric Gaussian targets and test all six ordered changes of favored target. Eight pilot seeds calibrate endpoint measurements without observing adaptation; 40 new seeds test bottleneck dimensions two and three. The response is the decrease in the pairwise allocation index for the original and destination targets, retaining all three reconstruction errors. Both capacities meet all eight mean endpoint criteria. Exchanging v reverses the ten-update route contrast, and exchanging its pattern at a common $\bar{a}_v$ changes the contrast at both capacities (Table 3). Reset equivalence holds at dimension two; at dimension three its interval crosses the $+0.02$ boundary and remains unresolved (Appendix K).

We then replace Gaussian targets with recorded temperature and humidity in a kitchen, living room, and laundry room from one house (Candanedo et al., 2017; Candanedo, 2017). Noise controls observation precision; shared learning reconstructs the recorded measurements. Chronologically separated blocks distinguish training, pilot readout/evaluation, and confirmation readout/evaluation. Room-specific calibration on eight pilot seeds fails to match endpoints, and confirmation on 40 new seeds also fails at both capacities. Nevertheless, with recipient weights fixed at exchange, changing v changes the response contrast, including at a common mean inverse scale. The swapped contrast is negative at dimension two but unresolved around zero at dimension three. This establishes a component effect on natural sensor measurements, while a matched-endpoint demonstration on natural data remains open (Appendix L).

6 Optimizer condition changes when the route difference appears

A small difference immediately after reset could still grow during continued learning. We therefore extend the opposing intervention to 500 updates under inherited Adam, fresh Adam, or momentum-free SGD. SGD's common learning rate is calibrated within each seed and target direction using first-batch inherited-Adam update norms and gradient norms, then held fixed; allocation responses are not used for calibration. All three continuations start from the Adam-formed route endpoints.

Inherited Adam produces its largest route differences after 10–20 updates. Fresh Adam gives 0.0096 at update 10 but 0.0274 at update 50; calibrated SGD gives 0.0123 at update 10, 0.0433 at update 50, and 0.0309 at

Table 3: Ten-update transfer across three-target tasks. All values are acquisition-minus-update response contrasts, except the final column, which subtracts the swapped-pattern contrast from the own-pattern contrast at common $\bar{a}_v$. Each dataset uses its own 40 seeds, shared across capacities. Endpoint matching passes only for the Gaussian task. Intervals and failed criteria appear in Appendices K–L.

Task, d_Z	Endpoints	Own v	Swapped v	Common-scale effect
Gaussian, 2	Matched	0.1945	-0.1005	0.0825
Gaussian, 3	Matched	0.4054	-0.2193	0.1541
Room sensors, 2	Unmatched	0.0601	-0.0254	0.0103
Room sensors, 3	Unmatched	0.1170	-0.0090	0.0257

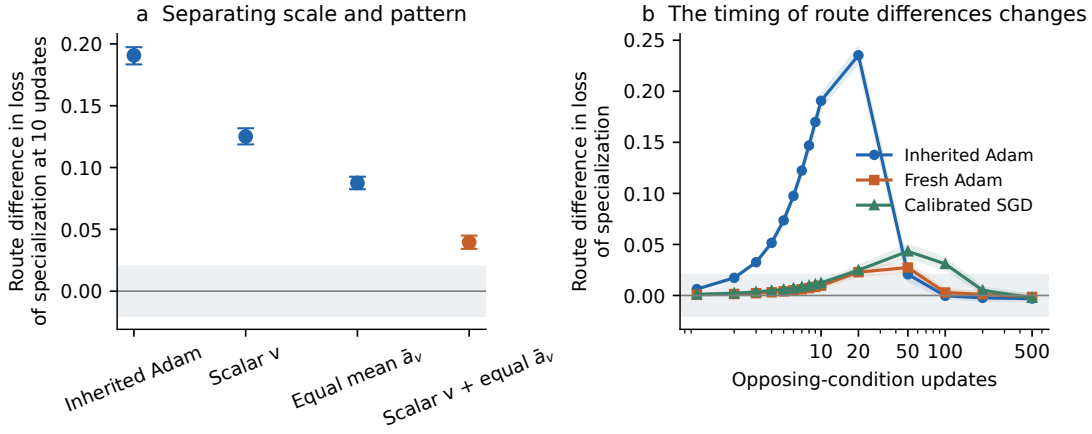


Figure 3: Scale/pattern interventions and observation horizon resolve different parts of the response. (a) Ten-update route contrast after retaining inherited state, scalarizing v, equalizing $\bar{a}_v$, or both. (b) Continued common opposition under inherited Adam, fresh Adam, or calibrated SGD. The band at ± 0.02 refers to the ten-update equivalence margin; later points have no pre-specified decision rule at that margin. Means and 95% paired seed-bootstrap intervals use 40 new seeds.

update 100 (Figure 3b). The update-50 intervals for fresh Adam and SGD lie above 0.02. That margin was fixed for the ten-update test and is only a reference for later points. The observations show delayed route differences after inherited Adam state has been removed, implicating differences retained on the model side without identifying their specific parameter or geometric source.

The route contrasts shrink toward zero by update 500. Under inherited Adam, the endpoint allocation difference is -0.00045 ($[-0.00146, 0.00057]$). This endpoint difference is distinct from the difference in change since formation, because the starting measurements were close rather than identical. The inherited-minus-fresh response contrast also changes sign by update 50, so the inherited trajectory is not described by adding a fixed nonnegative fast component to the fresh-Adam trajectory. Within this common opposing condition, optimizer state changes the timing and magnitude of reallocation; the large early difference does not imply a persistent allocation gap.

7 Implications and limits of the comparison

The result gives a concrete way to audit continued learning. Measure current specialization, apply the same future learning condition, and test which persistent components alter the response. In this construction, equivalent mean reconstruction measurements coexist with a large early contrast that can be reversed by exchanging v. Mean adaptive scale explains only part of that contrast. The equal-mean loss-weight control further shows that average task emphasis can conceal different retained states and responses. Together, these comparisons connect optimizer memory to the reallocation of information within a shared representation.

A concrete application is evaluating a checkpoint before resuming training on a changed task. The comparison suggests testing a common continuation with optimizer state inherited and reset, alongside current readout quality. In the present system these choices change the timing and magnitude of adaptation. Whether the same diagnostic improves decisions in fine-tuning requires validation on the target task, including its eventual performance.

Complementary experiments distinguish susceptibility from formation and maintenance. Current acquisition can redirect an already formed specialization under equal subsequent loss weights over 1,000 updates (Appendix B). Symmetrizing acquisition and updating reduces a formed bias, while designed state-following feedback can support long-lived regimes (Appendix C). A symmetric three-target construction reproduces acquisition-selected formation at two bottleneck dimensions after 600 updates (Appendix D). These controls establish that formation, current support, and response require different interventions. Three-target adaptation now supplies a separate test of susceptibility (Appendix K).

The matched adaptation evidence covers two or three Gaussian targets, small bottleneck networks, externally specified observation and loss weights, and finite horizons. The sensor extension covers one building and fixed chronological blocks; its seed intervals measure training/noise variability conditional on those records. Its unmatched endpoints prevent attributing the unmodified route contrast to a difference concealed by equivalent measurements. Establishing that comparison on natural data requires independently calibrated matching across richer histories and validation in other buildings or tasks. Coordinate exchange remains defined by common parameter ancestry, and equal global update norms still allow different directions and layerwise movement.

8 Conclusion

Similar measured specialization can conceal latent susceptibility to further learning in the studied two- and three-target Gaussian systems. Component exchange identifies inherited Adam second moments as a causal modifier of early reallocation, with effects beyond mean adaptive scale and, in the two-target control, beyond global update norm. The natural sensor extension retains component effects but fails endpoint matching, separating transfer of the intervention effect from transfer of the concealed-response claim. Reset equivalence and sign reversal depend on capacity, learning conditions, and horizon. Current readout quality, persistent training state, and subsequent learning conditions jointly determine what an adaptation comparison reveals.

Reproducibility statement

Code and generated data are available at `https://github.com/culturegeek/matched-specialization-different-adaptation` (v1.2.0). The archive contains source code, locally frozen plans, seed-level trajectories, checkpoints, and audit scripts for Experiments I–V-C, the IV-E follow-up, and the IV-F extensions. Code uses MIT; generated data and documentation use CC BY 4.0. The redistributed UCI sensor dataset retains its original CC BY 4.0 attribution. Appendix I maps claims to files and gives CPU-based integrity, numerical, figure-generation, and training-reproduction checks.

Broader impact statement

The work supplies controlled comparisons for examining shared representations whose input access and learning histories vary over time. Applications to deployed systems require direct validation of the interventions and measurements in those settings.

AI assistance statement

Generative AI systems were used extensively to propose most experiment designs and follow-up analyses, implement and execute experiments, analyze outputs, survey literature, and draft and translate the manuscript. The human author initiated the broader research program, commissioned this empirical study, authorized its

continuation at major decision points, reviewed the evidence and claims, and accepts responsibility for the paper and its public release.

References

Alessandro Achille, Matteo Rovere, and Stefano Soatto. Critical learning periods in deep networks. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=BkeStsCcKQ.

Kavosh Asadi, Rasool Fakoor, and Shoham Sabach. Resetting the optimizer in deep reinforcement learning: An empirical study. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Luis Candanedo. Appliances energy prediction. UCI Machine Learning Repository, 2017. URL https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction. CC BY 4.0.

Luis M. Candanedo, Véronique Feldheim, and Dominique Deramaix. Data driven prediction models of energy use of appliances in a low-energy house. *Energy and Buildings*, 140:81–97, 2017. doi: 10.1016/j.enbuild.2017.01.083.

Yifang Chen, Kevin Jamieson, and Simon Du. Active multi-task representation learning. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 3271–3298. PMLR, 2022.

Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 794–803. PMLR, 2018.

Michael C. Crair, Deda C. Gillespie, and Michael P. Stryker. The role of visual experience in the development of columns in cat visual cortex. *Science*, 279(5350):566–570, 1998. doi: 10.1126/science.279.5350.566.

Alexander D'Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226): 1–61, 2022. URL https://www.jmlr.org/papers/v23/20-1335.html.

Shibhansh Dohare, J. Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A. Rupam Mahmood, and Richard S. Sutton. Loss of plasticity in deep continual learning. *Nature*, 632:768–774, 2024. doi: 10.1038/s41586-024-07711-7.

Benjamin Ellis, Matthew T. Jackson, Andrei Lupu, Alexander D. Goldie, Mattie Fellows, Shimon Whiteson, and Jakob N. Foerster. Adam on local time: Addressing nonstationarity in RL with relative adam timesteps. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://arxiv.org/abs/2412.17113.

A. E. Harris, G. B. Ermentrout, and S. L. Small. A model of ocular dominance column development by competition for trophic factor. *Proceedings of the National Academy of Sciences*, 94(18):9944–9949, 1997. doi: 10.1073/pnas.94.18.9944.

Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang. Modality competition: What makes joint training of multi-modal network fail in deep learning? (Provably). In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 9226–9259. PMLR, 2022.

Stanislaw Jastrzebski, Maciej Szymczak, Stanislav Fort, Devansh Arpit, Jacek Tabor, Kyunghyun Cho, and Krzysztof Geras. The break-even point on optimization trajectories. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=r1g87C4KwB.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

O. M. Kiselev. A minimal bifurcation model of load imbalance in a softmax mixture-of-experts router. *arXiv preprint arXiv:2605.29121*, 2026. doi: 10.48550/arXiv.2605.29121.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, pp. 3519–3529, 2019.

Andrew Kyle Lampinen, Stephanie C. Y. Chan, and Katherine Hermann. Learned feature representations are biased by complexity, learning order, position, and more. *Transactions on Machine Learning Research*, 2024. URL `https://openreview.net/forum?id=aY2nsgE97a`.

Clare Lyle, Zeyu Zheng, Evgenii Nikishin, Bernardo Avila Pires, Razvan Pascanu, and Will Dabney. Understanding plasticity in neural networks. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 23190–23211. PMLR, 2023. URL `https://proceedings.mlr.press/v202/lyle23b.html`.

Mohammad Pezeshki, Oumar Kaba, Yoshua Bengio, Aaron C. Courville, Doina Precup, and Guillaume Lajoie. Gradient starvation: A learning proclivity in neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pp. 1256–1272, 2021.

Vasileios Sevetlidis and George Pavlidis. Training memory in deep neural networks: Mechanisms, evidence, and measurement gaps. *arXiv preprint arXiv:2601.21624*, 2026. URL `https://arxiv.org/abs/2601.21624`.

Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE National Convention Record*, 7(4):142–163, 1959.

Hajin Shim, Sung Ju Hwang, and Eunho Yang. Joint active feature acquisition and classification with variable-size set encoding. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Jack K. Wolf and Jacob Ziv. Transmission of noisy information to a noisy receiver with minimum distortion. *IEEE Transactions on Information Theory*, 16(4):406–411, 1970. doi: 10.1109/TIT.1970.1054469.

Qinyang Xu. Stored in optimizer state, valued by later training: A causal account of subliminal trait transfer. *arXiv preprint arXiv:2608.20442*, 2026. doi: 10.48550/arXiv.2608.20442. URL `https://arxiv.org/abs/2608.20442`.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, pp. 5824–5836, 2020.

Guide to the appendices

Appendix A gives the intervention details and estimates for the central adaptation comparison. Appendices B, C, and D examine how specialization forms, what supports it during continued learning, and how formation extends to three targets. Appendices E and F describe measurement checks and acquisition–compression calibration. Appendix G reports failed conditional predictors, while Appendix H reports factor controls, including initial failures and later tests with revised criteria; Appendix I distinguishes their statistical status and maps the evidence to reproduction files. Appendix J adds endpoint-prediction checks, fixed-protocol replications, and movement- and layer-controlled interventions. Appendices K and L test adaptation with three Gaussian targets and with natural sensor measurements.

A Endpoint and optimizer contrasts (IV-D–E)

IV-D uses 40 confirmatory seeds and two mirrored directions per seed. Mean aligned s is 0.3375 after acquisition-driven formation and 0.3439 after update-driven formation; corresponding target/non-target errors are 0.4034/0.8144 and 0.4000/0.8198. The route strengths are pilot-calibrated, not fitted to the confirmatory

endpoints. Neutral-continuation integral differences are -0.0025 ($[-0.0064, 0.0013]$) with Adam inherited and -0.0018 ($[-0.0068, 0.0030]$) after reset. Carried-minus-reset response interactions for acquisition, update, and combined pulses are 0.1657, 0.1390, and 0.1848, each exceeding the pre-specified 0.08 criterion.

The formation tolerances are ± 0.02 for aligned s, target and non-target errors, and their sum, and ± 0.01 for low-precision aligned s. IV-D uses response bands of ± 0.02 for neutral continuation and reset responses, and ± 0.05 for inherited-state pulses. IV-E uses ± 0.02 for the change under m exchange and for the reset response; the within-route update-history control uses ± 0.01 for its reset response. These criteria apply to intervals for mean paired differences.

IV-D classifies a contrast as equivalent when its 95% interval lies wholly inside the specified band, practically different when it lies wholly beyond either boundary, and unresolved otherwise. All three inherited-state pulse intervals lie wholly above $+0.05$ (Table 2), indicating positive, practically meaningful route differences. All reset-pulse intervals lie inside ± 0.02.

IV-D calibrates route strength on pilot seeds 8000–8015 and fixes the tolerances before seeds 8100–8139. IV-E uses pilot seeds 9000–9015 to set response criteria before seeds 9100–9139; the within-route control explicitly uses pilot seeds 9200–9215 to set its response criteria and equivalence margin before seeds 9300–9339. The plans document these operational choices but do not derive the margins from an independent practical-error criterion. Equivalence therefore denotes agreement at the stated resolutions in these synthetic comparisons.

Table 4: Combined-pulse response contrasts under component interventions. IV-E compares acquisition-formed minus update-formed routes; the control compares constant minus variable update-weight histories. Each uses 40 disjoint confirmatory seeds.

Component condition	IV-E route contrast	Equal-mean update-history control
Own state	0.1968 $[0.1910, 0.2029]$	0.0324 $[0.0294, 0.0354]$
Exchange m	0.1982 $[0.1912, 0.2054]$	0.0321 $[0.0277, 0.0362]$
Exchange v	-0.1265 $[-0.1348, -0.1183]$	-0.0141 $[-0.0176, -0.0110]$
Reset	0.0097 $[0.0078, 0.0117]$	0.0046 $[0.0031, 0.0062]$

In IV-E, the own-minus-m-exchange change is -0.0014 ($[-0.0044, 0.0018]$), inside ± 0.02. The acquisition-formed route's total second moment is 0.5577 times the update-formed route's, while the mean coordinate-wise inverse-second-moment scaling factor is 1.950 times larger. This scaling factor is not the actual parameter-update ratio: Adam's update also depends on the first moment, current gradient, and bias correction. Table 4 gives the main component contrasts.

The equal-mean update-history control fixes the target weight mean at 0.80 over 500 updates. Constant-minus-variable endpoint contrasts are aligned s: 0.0023 ($[-0.0024, 0.0072]$); target error: -0.0020 ($[-0.0043, 0.0002]$); non-target error: 0.0001 ($[-0.0042, 0.0043]$); total error: -0.0019 ($[-0.0039, 0.0002]$); and low-precision aligned s: 0.000756 ($[-0.000532, 0.002110]$). The second-moment ratio is 0.8215 ($[0.8140, 0.8283]$), and the inverse-scaling ratio is 1.1524 ($[1.1441, 1.1615]$). All pre-specified endpoint and response criteria pass. The inference concerns the joint change in weight variability and realized sequence, not an isolated order effect.

The prospective follow-up uses 40 new seeds (20000–20039). Adam uses $\eta = 0.002$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay 10^{-5}. With optimizer step count t and P parameters, define

$$\bar{a}_v = \frac{1}{P} \sum_{j=1}^{P} \frac{\eta}{\sqrt{v_j/(1-\beta_2^t)} + \epsilon}. \tag{3}$$

The acquisition-formed to update-formed ratio is 0.5539 ($[0.5350, 0.5732]$) for total v and 1.9042 ($[1.8710, 1.9387]$) for $\bar{a}_v$. These ratios are calculated separately for each seed and target direction, then averaged over directions within seed and over seeds. If v were uniform across coordinates, its total ratio would predict an inverse-scaling ratio of only 1.3513 ($[1.3282, 1.3745]$). Exploratory participation ratios, summarizing the effective number of coordinates carrying the second moment, are 19.17 and 38.93, respectively. Total second moment therefore does not reproduce the observed mean inverse scaling; coordinate inequality is associated with the discrepancy.

For common-scale interventions, the geometric mean of the two routes' $\bar{a}_v$ values is the target a_*. Pattern-preserving rescaling replaces every coordinate by cv_j, where positive c is found by 80 bisection iterations so that $\bar{a}_v(c) = a_*$. Pattern exchange first transplants the donor route's coordinate-wise v into the corresponding coordinates of the recipient model, then applies the same rescaling. Scalarization sets every coordinate to

$$v_j = \left(\frac{\eta}{a_*} - \epsilon\right)^2 (1 - \beta_2^t). \tag{4}$$

For scalarization at the recipient's own scale, a_* is its own $\bar{a}_v$ instead of the geometric mean. The recipient model weights, first moment, and step count remain fixed; achieved relative scale error is below 2×10^{-6}. These coordinate interventions are defined within models descending from the same checkpoint. Calibrated SGD uses zero momentum and the same weight decay. For each seed and favored target, its common learning rate is the geometric mean across routes of the first-batch inherited-Adam parameter-update norm divided by the current regularized-gradient norm; it then remains fixed (range 0.0194–0.0529, mean 0.03545). Allocation responses are not used in this calibration.

Table 5: Exploratory post-hoc recombination of second-moment mean scale and coordinate pattern. Values are acquisition-formed minus update-formed specialization loss after ten opposing updates, from the fixed follow-up CSVs.

Second-moment condition	Route difference [95% CI]
Own route scale and pattern	0.1907 $[0.1835, 0.1975]$
Own scale, scalar pattern	0.1251 $[0.1187, 0.1316]$
Common $\bar{a}_v$, own pattern	0.0875 $[0.0825, 0.0925]$
Common $\bar{a}_v$, scalar pattern	0.0395 $[0.0342, 0.0447]$

The post-hoc 2×2 interaction in Table 5 is 0.0176 ($[0.0151, 0.0198]$); the corresponding additive reconstruction reaches 90.8% of the observed mean. These combinations and this statistic were not in the fixed analysis plan, so they do not establish independent additive causes. In the prospective interventions, exchanging the other route's v pattern while equalizing $\bar{a}_v$ gives a route difference of -0.0110. A predictor fitted to the earlier seeds improves prediction of response *differences* on the new seeds, but a response-ratio predictor fails. The $\bar{a}_v$ ratio contains relative scaling but not the absolute denominator response, so the two findings are compatible.

Under continued opposition, route differences in specialization loss at updates 10, 20, 50, 100, 200, and 500 are, respectively: inherited Adam, 0.1907, 0.2353, 0.0207, -0.0003, -0.0023, -0.0031; fresh Adam, 0.0096, 0.0228, 0.0274, 0.0029, 0.0011, -0.0014; calibrated momentum-free SGD, 0.0123, 0.0248, 0.0433, 0.0309, 0.0050, -0.0022. At update 50, the fresh-Adam interval is $[0.0227, 0.0321]$ and the SGD interval is $[0.0368, 0.0499]$. At update 500, the endpoint allocation difference is -0.000454 ($[-0.001462, 0.000568]$), distinct from the drop difference of -0.003065 ($[-0.005642, -0.000316]$). In the separate ten-update-pulse-then-neutral trajectory, the allocation-index difference decreases from 0.06197 at update 50 to 0.00984 at update 510. The inherited-minus-fresh difference changes sign by update 50, so a fixed nonnegative fast component added to the fresh-Adam trajectory is inadequate; complete-state convergence remains untested.

A post-hoc update-path analysis reads only the fixed follow-up CSVs. Here the two target directions are first averaged within each route and seed; route ratios are then computed within seed and averaged across seeds. All four ratios below use this convention, which differs from averaging direction-specific ratios above. For inherited Adam, the acquisition-formed to update-formed ratios are 1.900 ($[1.867, 1.935]$) for $\bar{a}_v$, 2.028 ($[1.938, 2.116]$) for the first parameter displacement, 1.872 ($[1.844, 1.902]$) for cumulative ten-update parameter-path length, and 0.987 ($[0.978, 0.996]$) for cumulative gradient-norm path length.

For seed-level correlations and reconstruction, the $\bar{a}_v$ ratio averages direction-specific route ratios within each seed, using the convention whose overall mean is 1.9042. The parameter-path ratio instead divides the two route means after averaging target directions within each route and seed. The scaling ratio correlates 0.797 ($[0.644, 0.889]$) with parameter-path ratio; the between-route parameter-path difference correlates 0.806 ($[0.681, 0.889]$) with the allocation-response difference. Leave-one-seed-out linear reconstruction reduces mean squared error by 28% using the $\bar{a}_v$ ratio and 62% using realized path-length difference. In each fold, both the fitted model and the mean-only baseline use the other 39 seeds.

Path length is measured during the intervention, discards direction, and is not manipulated. It therefore describes a candidate mechanism linking scaling to response; quantifying a causal mediated effect requires controlling the realized movement. The reconstruction evaluates the measured ten-update response within these 40 seeds, rather than predicting it from formation endpoints alone.

B Formation and redirection controls (IV-A–C)

B.1 Input precision selects specialization under equal loss weights

We first ask whether a preference can develop without unequal loss weights. After 100 balanced initialization updates, each checkpoint is cloned into nine formation conditions: three acquisition conditions (A precise, balanced, or B precise) crossed with three update allocations (0.85/0.15, 0.50/0.50, or 0.15/0.85). Formation lasts 500 updates. Balanced acquisition contains exactly 250 updates with each target observed precisely, in randomized order; every condition receives the same total amount of high-precision information.

Under equal loss weights, precise acquisition of A yields mean $s = 0.3385$, while precise acquisition of B yields $s = -0.3338$. The paired contrast is 0.6723 ($[0.6660, 0.6794]$), above the pre-specified 0.30 criterion. Every confirmatory seed favors the precisely acquired target; balanced acquisition remains near zero. Because all conditions are evaluated using common inputs and newly fitted readouts, the difference persists beyond the momentary advantage of seeing a cleaner signal during training. Acquisition precision is sufficient to select specialization under symmetric explicit updating in this system.

The full factorial explains why this is not a fixed division of labor between information and learning (Figure 4a). Either control can favor a target. Agreement strengthens specialization, while conflict weakens it; at the tested levels, a strong opposing update weight can overcome the acquisition advantage. Both controls act through the gradients of the same network, so distinct interventions need not have orthogonal effects. These are 500-update outcomes, not estimates of limiting equilibria.

B.2 Acquisition also redirects already formed states

Does acquisition matter only before specialization develops? We formed A- and B-favored checkpoints by asymmetric updating while both targets were observed at noise standard deviation 0.15. We then fixed update weights at 0.50/0.50 and applied the same three acquisition conditions for 1,000 updates, with neutral checkpoints as an additional reference.

Opposite-target acquisition changes the favored target in all 40 seeds for both starting states (Figure 4b–d). The terminal A-precise minus B-precise contrast, averaged across formed states, is 0.7161 ($[0.7108, 0.7213]$), close to the neutral-start contrast of 0.7086. Balanced acquisition moves both formed states toward the center. The first zero crossings occur at mean recorded times of 16.9 and 16.8 updates, but trajectories can subsequently move nonmonotonically; these times do not measure settled adaptation. Current acquisition can redirect an update-formed representation under equal subsequent loss weights, within the observed 1,000-update horizon.

B.3 Strong one-sided updating changes the comparison

A separate formed-state experiment holds the update target strictly on one side and crosses it with precise acquisition of either the previously favored target or its alternative. Each formed checkpoint is cloned into four conditions for 400 updates. Updating only the reference target retains it as favored in 159/160 cells; updating only the alternative makes that target favored in all cells. Holding alternative-target updating fixed, however, acquiring that target precisely accelerates the trajectory relative to acquiring the reference target precisely. Under a common high-precision probe, the mean first-zero-crossing difference is 3.44 updates ($[2.81, 4.00]$). The larger threshold-specific estimate from the pre-specified protocol and its audit remain in Appendix E.

Terminal depth also depends on how acquisition and updating combine. In an exploratory analysis, aligned acquisition and update targets yield greater $|s|$ than crossed targets by 0.1411 under the common probe. The

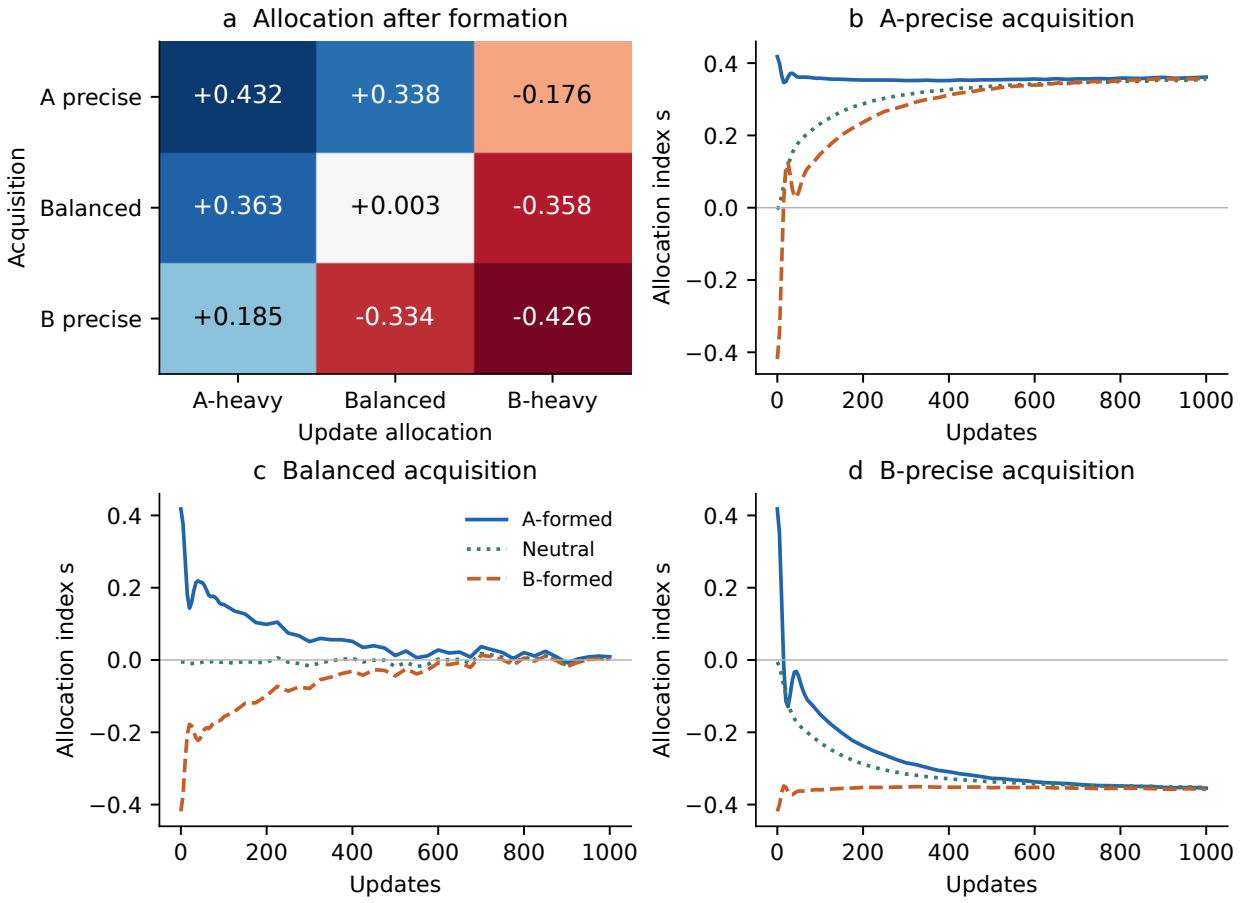


Figure 4: Current acquisition and updating jointly control specialization. (a) Mean allocation after 500 formation updates for the nine acquisition–update combinations. Equal update weights occupy the middle column. (b–d) With all subsequent update weights equal, acquisition redirects previously A-formed and B-formed states; a neutral starting state supplies a reference. Curves are seed means over 1,000 updates. Their early nonmonotonicity is why first zero crossing is distinguished from sustained switching or convergence.

acquisition main effect averages to approximately zero because its conditional effects have opposite signs under the two update targets. Thus the data do not support a general rule that acquisition controls only speed and updating controls only direction. Acquisition can select and redirect the favored target; sufficiently strong one-sided updating constrains the outcome while acquisition still changes the trajectory and terminal depth.

C Formation, current constraints, and feedback controls (II–III)

C.1 A formation condition need not remain as a lasting bias

Once a bias has formed, observing its persistence does not reveal what currently supports it. We therefore remove or change the supporting conditions. In the nonlinear two-target network, making both acquisition precision and update weights symmetric reduces mean $|s|$ from about 0.41 to 0.04 over 400 updates under the common probe. Decay continues through the final observation window. Because both controls are symmetrized, this intervention identifies their joint removal, not the necessity of biased updating alone.

A separate linear bottleneck experiment crosses conditions during formation and continued learning. Its constrained condition combines a one-dimensional representation with one-target updating; its unconstrained

condition uses two dimensions and updates both targets. Releasing the formation constraint reduces mean $|s|$ from 0.9706 to 0.1303, above the never-constrained baseline of 0.0592 but far below the formed bias. Further crossings distinguish capacity from update opportunity, as detailed below. Together, these interventions show that a formation history can leave a residual without maintaining its original magnitude under changed current conditions.

C.2 State-following feedback sustains long-lived regimes

To test a present sustaining mechanism, a related tanh network receives the complete input while a designed rule chooses which target is updated. The reinforcing rule follows the current allocation index: it preferentially updates the better-represented target. A fixed-propensity control freezes the same rule's probability at its initial value; an exploratory rule favors the worse-represented target. These interventions separate continued tracking of the learned state from an initial tendency to sample one target.

Three tests answer different questions. First, a strong imposed history difference nearly disappears under later common updating. Second, without an explicit history manipulation, state-following feedback amplifies incidental initial differences: terminal $|s|$ is 0.8683, versus 0.4207 under fixed propensity. Third, a separate mirrored-history experiment shows that state following and fixed propensity do not differ detectably in the signed effect of that history, but do differ in absolute specialization depth. The feedback result concerns amplification and support, not a uniquely stronger selection of the history-imposed sign. The detailed estimates below report the separate contrasts.

Capacity changes the maintained specialization. A three-dimensional bottleneck yields almost exclusive readout advantage for one target; a two-dimensional bottleneck yields two regimes with less extreme advantages for opposite targets. Weak opposing update pulses are followed by recovery, whereas stronger pulses can switch regimes. Without externally favoring either target, most trajectories retain their initial regime throughout the 8,000-update comparison. We use "metastable" operationally for this finite-time residence, recovery, switching, and coexistence. Counts, longer observations, and hysteresis tests are reported below; the evidence does not establish permanent bistability.

Asymmetric steady states have classical precedent in Harris et al. (1997), while Kiselev (2026) provides a recent minimal bifurcation construction based on positive feedback. Developmental formation and maintenance have also long been distinguished (Crair et al., 1998). The contribution here is their intervention-based realization and comparison in learned shared representations. The experiments identify present conditions that sustain a specialization, rather than assigning its persistence to history alone.

C.3 Detailed estimates

Experiment II uses a hidden-width-24 tanh encoder and linear heads, with 236 parameters at $d_Z = 2$ and 267 at $d_Z = 3$. One target is updated per mini-batch. Its principal reinforcement rule is $p_A(s_t) = 0.5 + \alpha \tanh(\beta s_t)$, with $\alpha = 0.40$ and $\beta = 1.5$; exploration changes the plus to minus. Balanced updating sets $p_A = 0.5$, and initial-fixed updating freezes $p_A(s_0)$.

The history-removal, no-history feedback, and mirrored-history tests are separate. In the first, a strong imposed history difference falls from 1.8822 to 0.0047 under common later learning (residual ratio 0.002); a weak common bias gives ratio 0.003. In the second, confirmatory seeds 300–339 have no imposed initial history: terminal $|s|$ is 0.1817 (balanced), 0.4207 (fixed), 0.8683 (state-following), and 0.0580 (exploratory). The state-following/fixed ratio is 2.064; the paired absolute-difference interval is $[0.368, 0.527]$.

In the third test, the first 100 updates use mirrored 55:45 and 45:55 sequences independent of current state. Pooling two disjoint 40-seed cohorts for this exploratory comparison, their immediate signed effect is 0.0811. Terminal signed effects are 0.2240 (fixed), 0.2170 (state-following), and 0.0015 (exploratory); the first two are not distinguishable. Their absolute deviations are 0.4503 and 0.8736, yielding ratio 1.94 and paired difference 0.4233 ($[0.3873, 0.4596]$). A mirrored 50:50 order control is consistent with zero. These contrasts support a depth effect without attributing a stronger signed history effect to state following.

The capacity/regime study yields terminal $|s| = 0.9842$ with $d_Z = 3$ and 0.3902 ($[0.3790, 0.4015]$) with $d_Z = 2$. Of 80 mirrored arms, 97.5% lie within 25% of the median branch magnitude, with 48.7% on each sign. A separate long run changes $|s|$ by -0.0004 ($[-0.0141, 0.0134]$) between 8,000 and 16,000 updates. Following 8,000 formation updates, opposite-target forcing for $K = 0, 800, 1600, 3200$ gives switching probabilities $0, 0.20, 0.425, 0.65$. Returning and switching trajectories reach 0.985 and 1.015 times the corresponding original/opposite magnitude.

At external update bias $h = 0$, positive and negative preparations retain their regimes in 39/40 and 40/40 cases for 8,000 updates. One unforced transition is observed. All 40 bias-sweep seeds switch and return, with mean threshold separation 0.1237 ($[0.1144, 0.1337]$). In ten diagnostic seeds, separation decreases from 0.2150 at 500 updates per bias level to 0.1575 at 2,000 and 0.1150 at 8,000. Residence, recovery, switching, and common-condition coexistence define the operational metastability claim; the dwell-time trend prevents inference of a nonzero quasistatic hysteresis width.

The linear constraint-release experiment in III uses independent target signals. With U denoting two dimensions/both targets updated and C one dimension/one target updated, the first letter indicates formation and the second continued measurement-phase learning. Terminal $|s|$ is 0.9862 in UC and 0.9856 in CC. CU falls from 0.9706 at formation to 0.1303, versus 0.0592 in UU. Further separation of the two controls gives 0.9037 for one dimension with both targets updated and 0.3368 for two dimensions with one target updated. This construction differs from the correlated nonlinear network in IV; it supplies a complementary constraint intervention, not another cell of the same factorial.

D Three-target formation estimates (V-A)

With two targets, every relative change lies on a single comparison axis. We therefore test the formation result using three symmetric three-dimensional Gaussian targets with pairwise correlation 0.35, mixed into nine input dimensions. The encoder has hidden width 32 and a bottleneck of dimension two or three. Update weights stay at 1/3. Precise acquisition favors A, B, or C throughout 600 updates; balanced acquisition allocates exactly 200 precise observations to each target. Each seed branches from the same initialized state. The three-dimensional bottleneck both matches the two-target compression ratio and equals one target's dimension, so it is not a pure control for compression ratio alone.

After formation, the encoder is frozen. Fresh linear readouts are fitted on unused probe samples and tested on separate held-out samples, with common high-precision inputs. We summarize relative errors with centered log-error scores,

$$q_i = \frac{1}{3} \sum_{j \in \{A,B,C\}} \log D_j - \log D_i, \qquad \sum_i q_i = 0. \tag{5}$$

A positive score indicates more precise readout than the geometric-mean error. Two coordinates describe this observable allocation plane. They are not the coordinates of the full internal representation, and their magnitudes are not directly comparable to the two-target index s.

Every target-specific acquisition condition favors its designated target in all 40 confirmatory seeds at both bottleneck dimensions. Relative to balanced acquisition, the designated target's allocation-score gain is approximately 0.37 with a two-dimensional bottleneck and 0.90–0.92 with three dimensions. The effects also spread across the allocation plane (Figure 5). To test whether this spread reduces to one empirical comparison axis, we summarize the three mean intervention-effect vectors by singular values. The second singular value measures the component beyond the best-fitting single axis: it is approximately 0.55 and 1.37 at the two capacities, with intervals excluding zero. Near-unit ratios of second to first singular values additionally verify the intended label symmetry. Exact estimates and intervals appear below. Precise acquisition also increases total reconstruction error relative to balanced acquisition: the result is a redistribution of readable information, not a uniform improvement for all targets.

This experiment extends acquisition-selected formation to three targets and establishes a second observable comparison direction. Appendix K tests subsequent adaptation and component exchanges. Persistence of three formed regimes under common conditions would require a separate maintenance intervention.

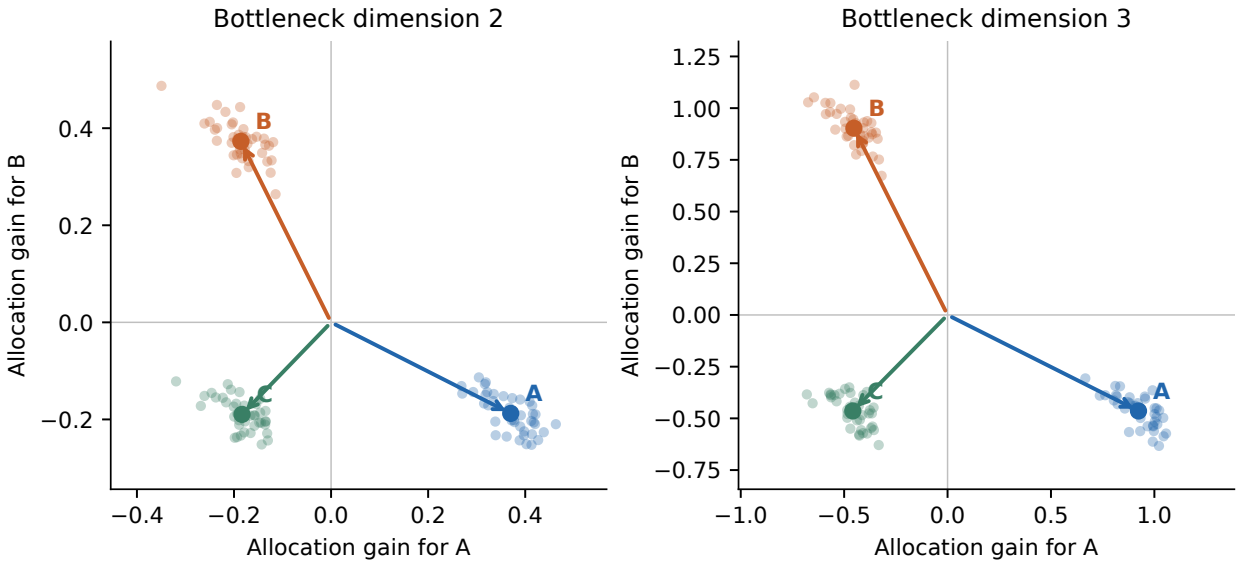


Figure 5: Three-target acquisition effects occupy an observable allocation plane. Each point is a seed's target-specific acquisition effect relative to its balanced-acquisition condition, with all update weights equal. Arrows summarize endpoint effects by connecting zero to the mean; trajectories are not shown. The coordinates are gains in q_A and q_B, with the change in q_C determined by their zero sum. Both bottleneck dimensions show three label-symmetric, noncollinear effects.

D.1 Detailed estimates

V-A uses 40 confirmatory seeds, each cloned across conditions at both capacities. The high-precision acquisition gain is the change in the designated target's centered log-error score relative to balanced acquisition. An update-positive control uses weights 0.8/0.1/0.1 with balanced acquisition. Table 6 gives ranges across target labels. All target-specific acquisition conditions favor the designated target in all seeds.

Table 6: Three-target formation after 600 updates. Gain and total-error ranges span the three target labels. Singular values summarize the target-specific intervention-effect matrix; reported values are bootstrap means.

d_Z	Acquisition gain	Update-control gain	σ_2	Total-error increase
2	0.3701–0.3731	0.4635–0.4668	0.5535	0.0749–0.0767
3	0.9030–0.9232	1.2239–1.2297	1.3656	0.1138–0.1148

The σ_2 intervals are $[0.5456, 0.5612]$ for $d_Z = 2$ and $[1.3491, 1.3815]$ for $d_Z = 3$. The ratios σ_2/σ_1 are 0.9839 ($[0.9725, 0.9947]$) and 0.9876 ($[0.9767, 0.9971]$), above the pre-specified 0.70 criterion. Nonzero second contrast and near-label symmetry are different claims; neither identifies the dimension of the internal learning state. In an exploratory normalization against the best linear-code distortion floors (0.3344 and 0.0025), acquisition removes 63.23% and 63.67% of removable distortion, and the update control removes 75.61% and 75.39%. These similarities do not constitute a confirmatory capacity-invariance test.

E Network, acquisition, and measurement details (IV)

The central IV network has the structure $6 \to 32 \to 2$ with tanh after each encoder layer, followed by linear reconstruction heads. In IV-B, training/probe/test sample counts are 3,072/768/1,536; batch size is 256, learning rate 0.002, weight decay 10^{-5}, and post-hoc readout ridge coefficient 10^{-3}. Noisy target channels are divided by $\sqrt{1+\sigma^2}$ before the orthogonal mixing. Training targets are noise-free. IV-B uses 100 initialization

and 500 formation updates; IV-C continues formed states for 1,000 updates. These models are nonlinear shared encoders even though their training and post-hoc readout heads are linear.

IV-A follows 40 seeds with two mirrored formed states through four acquisition–update cells for 400 updates. Under the original probe, crossing the pre-specified aligned threshold -0.08 takes 40.6 updates with reference acquisition versus 22.2 with alternative acquisition, holding alternative-target updating fixed. The difference is 18.4 ($[13.2, 25.8]$). Remeasuring stored trajectories with common low-noise inputs expands the allocation range from approximately $[-0.154, 0.160]$ to $[-0.451, 0.462]$. The original-input audit rows are bit-identical to the original trajectories; this is a measurement addition, not a new learning run.

First-zero-crossing differences are 3.50 under the original probe and 3.44 ($[2.81, 4.00]$) under the common probe. Changing the original threshold from 0.04 to 0.06 to 0.08 changes the time difference from 6.50 to 8.44 to 18.38. Thresholds at or above 0.10 increasingly produce non-arrivals and sensitivity to right censoring. The pre-specified 18.4 estimate is therefore threshold-specific; agreement of zero-crossing and continuous-trajectory effects supports the direction of the acquisition effect across measurements.

The exploratory terminal-depth alignment contrast is 0.0376 with the original probe and 0.1411 ($[0.1370, 0.1455]$) with the common probe. The common-probe acquisition main effect is 0.0002 ($[-0.0011, 0.0014]$); conditional effects reverse sign, $+0.1413$ under reference updating and -0.1410 under alternative updating. Symmetric acquisition/update continuation reduces original-probe $|s|$ from 0.1326 to 0.0146 and common-probe $|s|$ from 0.4067 to 0.0413, with mean residual ratio 0.1009. These outcomes distinguish terminal depth, trajectory, and maintenance rather than assigning all three to an acquisition main effect.

IV-B's common low-precision probe retains the acquisition contrast at 0.2237. IV-C retains it at 0.2344 ($[0.2322, 0.2365]$), compared with the common high-precision formed-start contrast of 0.7161. These cross-probe agreements support the favored-target comparison while demonstrating why allocation-index magnitudes should not be treated as measurement-invariant physical quantities.

F Acquisition and compression calibration (Experiment I)

The acquisition control is calibrated in a linear Gaussian setting where observation loss and compression loss can be distinguished. The decomposition follows indirect source coding (Shannon, 1959; Wolf & Ziv, 1970). Under squared error, for a target X, acquired data Y, and a predictor $g(Y)$, conditional-expectation orthogonality gives

$$\mathbb{E}\|X - g(Y)\|^2 = \mathbb{E}\|X - \mathbb{E}[X \mid Y]\|^2 + \mathbb{E}\|\mathbb{E}[X \mid Y] - g(Y)\|^2. \tag{6}$$

The first term cannot be reduced merely by expanding a code constructed from the same Y. A representation constraint can further increase the second term.

For two correlated three-dimensional targets mixed by a seed-specific random orthogonal transformation, we enumerate channel subsets at observation budgets $B_{\text{obs}} \in \{2, 3, 4, 6\}$ and fit optimal rank-d_Z linear representations for $d_Z \in \{1, \ldots, 6\}$. There are 8,192 training and 4,096 test samples per seed, across 20 seeds. The unconstrained mean distortion is 0.0004; acquisition-only and representation-only constraints increase it by 0.3956 and 0.3262, respectively. These are calibration contrasts from the acquisition–representation grid. The best two-channel subsets under the two target weightings have mean Jaccard overlap 0.0167, versus 0.2444 for random subsets. This is a manipulation check for task-dependent acquisition, not an identity forced by the target labels after mixing.

A separate analytic check uses a rank-three encoder, for which balanced population error is $D_A = D_B = (1 - \rho)/2$. Across six correlations, gradient training with optimal readout matches this curve with RMSE 0.0030 and maximum error 0.0042. Expanding from three to six dimensions improves distortion by 0.3236 in this separate calibration, not the 0.3262 grid contrast.

The sequential test supplies four observations, each of A or B, giving 16 fixed schedules. A noisy initial cue predicts the later target demand; a cue-dependent policy chooses a complete schedule once, rather than repeatedly tracking an internal learned state. Selection uses 30,000 training and 10,000 validation samples, followed by 20,000 independent test samples. The best fixed objective is 0.181864; the cue-dependent objective

is 0.124223, a 31.7% reduction. Shuffling the cue yields 0.181039 with the same policy class. The gain therefore depends on cue information rather than the larger class alone.

Temporal dependence also couples neighboring schedules. If $L_s(W)$ and $L_t(W)$ are endpoint-policy losses, the shared-decoder excess

$$B(\alpha) = \min_W\{(1-\alpha)L_s(W) + \alpha L_t(W)\} - (1-\alpha)\min_W L_s(W) - \alpha \min_W L_t(W) \tag{7}$$

is nonnegative for $0 \le \alpha \le 1$ by construction. For an interior mixture and attained minima, zero excess requires a common minimizing decoder. The measured result is the frequency with which this common-optimum condition fails: positive excess occurs on 0/32 neighboring paths at temporal correlation $\phi = 0$, 4/32 at 0.4, and 18/32 at 0.8. The maximum is 0.039801, and dedicated decoders remove the excess. At $\phi = 0.8$, the exact population policy objective has two local optima; 13/20 jointly trained seeds reach the oracle schedule.

G A candidate coordinate fails conditional prediction (III)

A separate test asks whether a slowly changing variable predicts switches between maintained specializations. An externally specified loop couples representation, a framing variable κ_t that sets task emphasis, and evaluation sensitivity. Eighty arms form history-consistent regimes with mean $|s| = 0.9854$. No spontaneous switch occurs in 8,000 unperturbed updates; the 90th-percentile weak-perturbation recovery time is 200 updates, giving an observed time-scale-ratio lower bound of 40. Independent reconfirmation with opposing forcing lasting $K = 0, 1600, 3200, 4000, 4800, 5600, 6400$ updates gives rounded switch probabilities $0, 0, 0.138, 0.450, 0.688, 0.838, 0.938$.

The original, disjoint coordinate-audit sample has rates 0.4625, 0.6250, and 0.8750 at $K = 4000, 4800, 5600$. AUC measures how well a predictor ranks switching above non-switching trajectories. The pooled κ AUC is 0.712, versus 0.600 for s. Within those intervention levels, the κ AUCs are 0.565, 0.567, and 0.380. Seed-separated five-fold validation changes AUC by 0.0049 when κ is added to K and s, while log loss worsens slightly. The two sets of switch rates therefore describe different samples, not conflicting estimates from one CSV. An earlier II head-distance candidate also fails an independent conditional test (AUC 0.373 at $K = 1600$). These are failures of the tested predictors, not proofs that no compact dynamical coordinate exists.

H Supporting finite-state factor controls (III)

A separate resource-management environment tests whether task framing, causal prediction, prediction horizon, evaluation horizon, risk sensitivity, and stopping can be manipulated distinctly. A controller supplies current demand from a five-level resource while maintaining equipment against future failure. Resource level, binary demand, and binary equipment state give 20 external states. Standard sensors have accuracy 0.62; an additional target-specific sensor has accuracy 0.94. A ten-step dynamic-programming oracle supplies policy and Pareto references, with a tradeoff environment and a null environment admitting a common optimum.

The initial confirmatory run passes framing and prediction-horizon checks but fails causal-model, evaluation-horizon, and risk-sensitivity criteria. The transition intervention is too small, the evaluation effect misses its threshold, and the risk threshold exceeds baseline risk. Revised strengths and attainable relative criteria are fixed before 40 new seeds, where all five families pass. These later successes do not change the status of the initial failures. A reframing factorial switches only when long prediction, long evaluation, and frame plasticity jointly occur under the designed rule; better alternative information improves frame–equipment agreement rather than simply increasing switch counts. These controls support the infrastructure for manipulating the specified factors separately.

I Statistical status and provenance

Confirmatory outcomes retain their pre-specified observation windows, contrasts, and margins. Terminal-depth alignment in IV-A and removable-distortion normalization in V-A are exploratory. Threshold sensitivity and conditional predictor tests are audits, not replacements that retroactively redefine the primary outcomes. Within-seed pairing is preserved; mirrored labels are not independent seeds. Failed initial factor checks and failed conditional predictors remain reported in Appendices H and G.

The question-based sections map to the reproduction records as follows: formation and redirection use IV-B, IV-C, and IV-A; current support uses II, III's constraint-release construction, and IV-A neutral continuation; latent susceptibility uses IV-D, IV-E, the IV-E update-weight control, and the IV-E follow-up; the conditional coordinate audit uses III-C; three-target formation uses V-A. Experiment I supplies acquisition–compression calibration. The follow-up's fixed intervention tests and its later post-hoc recombination are stored and labeled separately. The constructions provide complementary tests under their separately specified parameters.

The archive's `docs/PAPER_TO_DATA.md` links each figure and principal table to its input files. From the archive root, `python scripts/verify_data.py` checks preserved-file hashes and the follow-up's frozen plans and results; `python scripts/check_core_results.py` independently recomputes 27 central means from stored CSVs. After installing `requirements-analysis.txt`, `python scripts/make_figures.py` regenerates the five English figure families. The numerical check covers the selected means. A separate interval ledger links every printed confidence interval to a saved result and its rounding precision. The pooled mirrored-history depth contrast and the two IV-A terminal-depth contrasts use 20,000 paired seed-bootstrap resamples, with mirrored labels averaged within seed. Their reproducible exploratory calculations are provided by `scripts/audit_manuscript_intervals.py`; registered experiment results and decision criteria remain separately identified.

Training reruns are described in `docs/REPRODUCING.md`, including a one-seed trajectory comparison and the full 40-seed follow-up. The recorded follow-up environment is Python 3.11.7, NumPy 2.4.6, and PyTorch 2.13.0 on CPU with one thread. The environment and scope of the separate package checks are recorded in `provenance/validation.json`; they do not constitute a rerun of every historical experiment. Reruns write to new output directories so that the archived confirmatory data, failed criteria, and exploratory records remain available for comparison.

The IV-F files are in `experiments/iv-extension/`. The two registration files record local pre-run hashes. Its `README.md` maps all modes, predictions, layer summaries, and failed criteria to data and rerun commands. `audit_extension.py` independently checks raw-CSV means, completeness, shared starts, norm constraints, and frozen hashes. Before new training, the baseline checkpoint and six pulse modes reproduced saved seed 20000 exactly.

J Endpoint prediction, transfer, and mechanism extensions (IV-F)

The extension asks which parts of the short response contrast are predictable from saved endpoints, transfer to altered networks and targets, and survive control of movement. Existing-data analyses use three disjoint 40-seed cohorts. New training uses the same 32 previously unused seeds (32000–32031) across nine settings, comprising four architectural/target settings and a 3×2 second-moment-decay/formation-duration grid with one shared baseline. Layer and movement interventions reuse 40 saved follow-up seed pairs. The protocol and code were locally frozen before these new runs; the existing-data analyses remain exploratory. Mirrored directions are averaged within seed. Intervals use 5,000 paired seed-bootstrap resamples and are individual 95% intervals.

J.1 Do residual endpoint differences predict the response?

Within each cohort, we predict the combined-pulse response contrast from the paired endpoint differences in target error, other-target error, and low-precision aligned s. Ordinary least squares uses training-fold standardization and a held-out-seed evaluation. Its baseline predicts the mean of the other 39 seeds. Error ratios near or above one indicate little predictive benefit (Table 7). Applying the IV-D coefficients unchanged

to IV-E and the follow-up gives error ratios 1.032 and 0.999 relative to the IV-D mean predictor. The intercept estimates the contrast at zero differences in the three measured features; it is a conditional linear description, not a causal adjustment for unmeasured geometry.

Table 7: Exploratory endpoint prediction. MSE ratios compare held-out errors with the training-mean baseline; the last column is the conditional intercept and its seed-bootstrap interval.

Cohort	MSE ratio	Contrast at zero endpoint differences
IV-D	1.118	$0.1935\ [0.1841, 0.2023]$
IV-E	1.023	$0.2001\ [0.1915, 0.2081]$
IV-E follow-up	0.984	$0.1874\ [0.1776, 0.1965]$

J.2 Transfer under an unchanged formation protocol

The four settings retain two three-dimensional targets, a two-dimensional bottleneck, 100 initial updates, 500 formation updates, target loss weight 0.80 in the update route, and the common ten-update pulse. Width 64 changes only the hidden width; ReLU replaces both encoder tanh activations; $\rho = 0$ removes target correlation. No response-dependent recalibration is performed. All five original endpoint tolerances pass at every setting. Own, m exchange, v exchange, reset, common $\bar{a}_v$, and common $\bar{a}_v$ with exchanged v are evaluated at every setting.

Table 8 shows positive inherited-state contrasts and reversal after v exchange in every setting. At common $\bar{a}_v$, the paired exchange effects are 0.1028, 0.1568, 0.0491, and 0.0137 for baseline, width 64, ReLU, and independent targets, respectively; every interval lies above zero. The reset interval at width 64 lies wholly above 0.02, failing the fixed equivalence criterion. Thus the component effect transfers across these changes, while reset equivalence does not transfer to every setting.

Table 8: Fixed-protocol replications on the same 32 new seeds. All settings pass endpoint matching. Entries are ten-update route contrasts and paired 95% intervals.

Setting	Own state	Exchange v	Reset
Baseline	$0.1962\ [0.1881, 0.2043]$	$-0.1290\ [-0.1379, -0.1198]$	$0.0073\ [0.0054, 0.0092]$
Width 64	$0.2877\ [0.2788, 0.2962]$	$-0.1502\ [-0.1606, -0.1395]$	$0.0249\ [0.0217, 0.0282]$
ReLU	$0.0879\ [0.0739, 0.1003]$	$-0.0831\ [-0.0921, -0.0731]$	$-0.0026\ [-0.0045, -0.0010]$
$\rho = 0$	$0.0668\ [0.0571, 0.0770]$	$-0.0901\ [-0.1001, -0.0801]$	$0.0056\ [0.0020, 0.0097]$

J.3 Second-moment decay and formation duration

The grid uses $\beta_2 \in \{0.9, 0.99, 0.999\}$ and 500 or 1,500 formation updates, with the same decay parameter during initialization, formation, and the pulse and with Adam bias correction retained. All six settings pass endpoint tolerances. At 500 updates the own-state contrast grows from 0.0298 through 0.1385 to 0.1962 as β_2 increases. The smallest fails the fixed lower-bound criterion of 0.05. Exchange reverses the contrast at 0.99 and 0.999 after 500 updates, but only at 0.999 after 1,500 updates. Common-scale exchange effects remain positive throughout (Table 9).

All three reset conditions after 1,500 formation updates fail the ± 0.02 criterion, as does $\beta_2 = 0.9$ after 500 updates. The grid therefore supports a decay- and duration-dependent component effect while narrowing the conditions for reversal and reset equivalence. Because β_2 changes the formation trajectory and current adaptation as well as retained history, this is not an isolated manipulation of memory duration.

J.4 Where does exchange act, and does it survive equal movement?

The saved follow-up checkpoints show that total v and mean inverse scaling describe different distributions. The first encoder layer has 224 parameters and contains 5.9% of total v in the acquisition route versus 13.0%

Table 9: Decay and duration grid, 32 paired seeds per setting. Pattern effect is the route-contrast difference before minus after v exchange at common $\bar{a}_v$. All settings satisfy endpoint tolerances.

β_2	Formation	Own state	Exchange v	Pattern effect
0.9	500	0.0298 [0.0278, 0.0318]	0.0104 [0.0082, 0.0127]	0.0058 [0.0053, 0.0064]
0.9	1500	0.0764 [0.0736, 0.0794]	0.0494 [0.0462, 0.0526]	0.0120 [0.0111, 0.0131]
0.99	500	0.1385 [0.1329, 0.1444]	−0.0720 [−0.0770, −0.0672]	0.0673 [0.0625, 0.0721]
0.99	1500	0.2859 [0.2789, 0.2925]	0.0297 [0.0204, 0.0390]	0.1334 [0.1204, 0.1448]
0.999	500	0.1962 [0.1881, 0.2043]	−0.1290 [−0.1379, −0.1198]	0.1028 [0.0947, 0.1109]
0.999	1500	0.4341 [0.4277, 0.4408]	−0.1279 [−0.1416, −0.1139]	0.2528 [0.2414, 0.2644]

in the update route; its mean inverse scaling is 1.130 versus 0.591. The two heads together contain 66.2% versus 43.9% of total v. Nevertheless, exchanging only head v changes the response contrast by less than 0.001, whereas exchanging either encoder layer changes it substantially (Table 10). Layers include weights and biases. The intervention results are non-additive and do not turn these total-mass shares into causal contribution fractions.

To control realized movement, each of the four branches (two routes with own or exchanged v) proposes an Adam update from its current state. The geometric mean of the four proposed Euclidean norms becomes their shared target at that step. Each vector is rescaled without changing direction; m, v, and step count continue to update using gradients at the actual positions. The maximum relative discrepancy from the target norm is below 2×10^{-6}. The resulting exchange effect is 0.0231 [0.0208, 0.0253]. Thus a component effect remains when global movement is equal. Directions and layerwise movements remain free to differ; the joint control rule does not identify a natural-trajectory mediation fraction.

Table 10: Layer and movement interventions on 40 saved seed pairs. Layer rows exchange only the named layer's v; equal-movement rows apply the joint norm-control rule.

Intervention	Route contrast [95% CI]
Own	0.1907 [0.1837, 0.1976]
First encoder layer	0.0115 [0.0060, 0.0171]
Second encoder layer	0.0611 [0.0558, 0.0665]
Both heads	0.1911 [0.1838, 0.1978]
All parameters	−0.1183 [−0.1264, −0.1099]
Equal movement, own	0.0303 [0.0269, 0.0338]
Equal movement, exchanged	0.0072 [0.0040, 0.0104]

K Matched three-target adaptation (V-B)

To test whether optimizer-mediated reallocation survives a third competing target, we retain the V-A Gaussian construction: three three-dimensional targets with pairwise correlation 0.35, a $9 \to 32 \to d_Z$ tanh encoder, and three linear heads. For each $d_Z \in \{2, 3\}$ and original target i, acquisition formation observes i precisely with equal loss weights. Update formation balances precision across targets while assigning i weight w and the others $(1 - w)/2$. Both descend from the same 100-update balanced ancestor and train for 600 further updates with Adam. Learning rate, decay, batch size, and train/probe/test counts are 0.002, $(0.9, 0.999)$, 256, and 3072/768/1536; weight decay is 10^{-5} and $\epsilon = 10^{-8}$.

Eight pilot seeds (33000–33007) compare $w = 0.55, 0.60, \ldots, 0.95$ using endpoints alone. For each capacity we minimize the largest absolute mean endpoint difference divided by its tolerance. The selected weight is 0.65 at both capacities. Eight measurements align the original target and its two cyclic alternatives: three errors, their sum, two pairwise indices, and the two indices under a low-precision probe. Tolerances are ±0.02, or ±0.01 for low-precision indices. The three original-target labels are averaged within seed in this symmetric construction. Confirmation on seeds 34000–34039 meets every interval-based endpoint criterion at both capacities. These are operational measurement tolerances, not externally validated performance thresholds.

Each endpoint receives two destination-target pulses, giving all six ordered directions. A pulse observes destination j precisely and weights its loss 0.8, with 0.1 for each other target. Define $s_{ij} = (D_j - D_i)/(D_j + D_i)$ and response $s_{ij}(0) - s_{ij}(t)$. The route contrast subtracts update-formed response from acquisition-formed response. All three errors and centered log-error scores are retained. The six optimizer modes use own state, exchanged m, exchanged v, fresh Adam, own v at common $\bar{a}_v$, or exchanged v at the same common scale. Exchanges preserve recipient weights, the other moment, and step count; scale normalization uses the geometric mean of the two original scales and 80 bisection iterations including ϵ.

The protocol fixes ten updates as primary and 50 as secondary, with measurements at 0, 1, 5, 10, 20, and 50 updates. All six directions are averaged within seed; the same 40 seeds are reused across capacities, rather than counted as 80 independent seeds. Individual 95% intervals use 5,000 paired-seed bootstrap resamples. Plans, calibration, analysis, and code were locally frozen before confirmation; all seeds and failed criteria are retained. A saved V-A seed reproduces exactly before this extension is run.

Both capacities show a positive own-state contrast and a negative swapped-v contrast (Table 11); the six direction-specific means have these same signs. First-moment changes lie within ± 0.02. At common mean scale the pattern effect remains positive, although the swapped contrasts themselves remain positive: component sensitivity transfers more broadly than reversal under every scaling condition. Reset meets ± 0.02 at dimension two; its dimension-three interval crosses 0.02, so equivalence is unresolved rather than disproved by an interval wholly outside the margin.

Table 11: Three-target Gaussian response contrasts at ten updates. Each column uses the same 40 seeds. The last two rows are paired differences between intervention contrasts; all intervals are individual 95% seed-bootstrap intervals.

Intervention	$d_Z = 2$	$d_Z = 3$
Own state	$0.1945\ [0.1906, 0.1987]$	$0.4054\ [0.3951, 0.4152]$
m exchanged	$0.1943\ [0.1899, 0.1988]$	$0.4063\ [0.3962, 0.4161]$
v exchanged	$-0.1005\ [-0.1054, -0.0955]$	$-0.2193\ [-0.2266, -0.2118]$
Adam reset	$0.0116\ [0.0101, 0.0132]$	$0.0203\ [0.0178, 0.0226]$
Own pattern, common scale	$0.0904\ [0.0864, 0.0947]$	$0.1749\ [0.1668, 0.1822]$
Swapped pattern, common scale	$0.0079\ [0.0037, 0.0121]$	$0.0208\ [0.0144, 0.0273]$
v exchange effect	$0.2950\ [0.2901, 0.3003]$	$0.6247\ [0.6129, 0.6371]$
Common-scale pattern effect	$0.0825\ [0.0792, 0.0860]$	$0.1541\ [0.1479, 0.1597]$

At 50 updates, own-state contrasts are approximately 0.0000 and 0.0069 at dimensions two and three; reset contrasts are 0.0255 and 0.0596. The own-minus-swapped-v effects are then -0.0132 and -0.0263, and the common-scale effects are -0.0241 and -0.0479. Thus the early positive exchange effect does not persist with the same sign. The primary result extends the matched comparison to a third Gaussian target over ten updates; it does not supply a horizon-independent equivalence or a general three-target transition model.

L Natural three-room sensor extension (V-C)

We test natural cross-target and temporal structure using the UCI Appliances Energy Prediction data (Candanedo et al., 2017; Candanedo, 2017). The source contains 19,735 ten-minute records from one house, January–May 2016. The targets are kitchen temperature/humidity (T1, RH_1), living-room measurements (T2, RH_2), and laundry-room measurements (T3, RH_3). The task reconstructs these recorded values after controlled observation noise; it is neither future forecasting nor the source dataset's appliance-energy benchmark.

The first 9,867 rows train the model. Four subsequent chronological blocks fit and evaluate pilot readouts (2,816 and 2,817 rows), then fit and evaluate confirmation readouts (1,829 and 1,830 rows). We omit 144 rows at each boundary. Training ends on March 20, pilot evaluation on April 30, and confirmation evaluation spans May 15–27. Means and standard deviations come only from the training block and are fixed thereafter. D_i averages squared error over the room's two training-standardized variables, without normalization by

evaluation-block variance. The model's formation and pulse updates use only the training block; fresh readouts use their designated later blocks. Gaps reduce immediate overlap but do not establish temporal independence.

The encoder becomes $6 \to 32 \to d_Z$ with two-dimensional heads; training settings, precision noise, six directions, six modes, and horizons otherwise follow V-B. Room labels keep their physical meaning. Eight pilot seeds (35000–35007) select a weight separately for each original room and capacity, among 1/3, 0.40, 0.45, ..., 0.95. Selection minimizes the same largest standardized mean endpoint discrepancy, without using pulse responses or confirmation blocks. Selected weights for kitchen/living/laundry are 0.40/0.95/0.75 at dimension two and 0.95/0.95/0.90 at dimension three. Even the best candidates exceed the largest allowed mean discrepancy by factors of 5.30–12.44. The fixed protocol carries these candidates into confirmation without expanding the search.

Forty new confirmation seeds (36000–36039) again fail endpoint matching. Because rooms are not exchangeable, matching requires all eight criteria for each original room, rather than averaging away room-specific differences. Only 1 of 24 criteria passes at dimension two, and none passes at dimension three. Direct input readouts verify that precise acquisition reduces the designated room's error: gains are 0.1784, 0.4822, and 0.1824 for kitchen, living room, and laundry room. Thus observation control remains effective, while this grid and formation schedule do not yield matched route endpoints.

Exchanging v changes the ten-update response contrast at both capacities, including when its mean inverse scale is held common (Table 12). First-moment changes and reset contrasts lie within ± 0.02. The inherited contrast exceeds the $+0.05$ criterion only at dimension three; the swapped contrast excludes zero negatively only at dimension two. These distinct criteria prevent interpreting a positive exchange effect as universal sign reversal. Since the endpoints are unmatched, the exchange comparison identifies an intervention on each recipient's retained state, but the original route difference cannot be described as hidden by equivalent endpoint measurements.

Table 12: Natural sensor response contrasts at ten updates; endpoints are unmatched. Each column uses the same 40 seeds. The last two rows are paired differences between intervention contrasts; all intervals are individual 95% seed-bootstrap intervals.

Intervention	$d_Z = 2$	$d_Z = 3$
Own state	$0.0601\ [0.0456, 0.0741]$	$0.1170\ [0.1024, 0.1320]$
m exchanged	$0.0620\ [0.0472, 0.0765]$	$0.1157\ [0.1007, 0.1310]$
v exchanged	$-0.0254\ [-0.0381, -0.0130]$	$-0.0090\ [-0.0232, 0.0049]$
Adam reset	$-0.0076\ [-0.0168, 0.0017]$	$0.0046\ [-0.0062, 0.0152]$
Own pattern, common scale	$0.0230\ [0.0087, 0.0369]$	$0.0694\ [0.0561, 0.0831]$
Swapped pattern, common scale	$0.0127\ [-0.0006, 0.0257]$	$0.0437\ [0.0296, 0.0578]$
v exchange effect	$0.0854\ [0.0772, 0.0934]$	$0.1261\ [0.1122, 0.1410]$
Common-scale pattern effect	$0.0103\ [0.0064, 0.0140]$	$0.0257\ [0.0210, 0.0306]$

At 50 updates, the own-state contrasts are 0.0842 and 0.1156; reset contrasts are 0.0124 and 0.0392. The ten-update reduction therefore does not imply enduring route equivalence. All intervals resample training/noise seeds conditional on one building and fixed time blocks, and do not estimate between-building or between-period uncertainty. This extension supports a component effect during controlled reconstruction of natural measurements. A natural matched-endpoint claim requires new calibration data and confirmation across additional records, while preserving the unsuccessful comparison reported here.

The archive directories `experiments/v-three-adaptation/` and `experiments/v-natural/` retain all plans, pilot/confirmation cells, checkpoints, frozen manifests, and rerun instructions. Independent CSV audits pass 1,603 and 1,859 checks, including bootstrap estimates, intervention starts, scaling constraints, and hashes; V-C also verifies temporal separation and training-only standardization. Runs use Python 3.12.14, NumPy 2.4.6, PyTorch 2.13.0, and single-threaded CPU workers. Local freezing precedes confirmation but is not public preregistration.